A Validated Leakage Audit for Wearable Physiological Regression, with Application to PPG-Based Glucose and Blood-Pressure Estimation

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Abstract*—*Objective: **Reported accuracies of cuffless physiological estimation from photoplethysmography (PPG) are often inflated by data leakage that standard subject-wise splitting does not detect. We develop and validate an audit that detects, measures and bounds such leakage.** ***Methods:*** **The audit comprises (i) a detector for label-construction leaks (labels replicated across signal rows when public tables are built), (ii) machine-checked split protocols, (iii) a leakage ladder that measures how much each mechanism inflates a metric, and (iv) a null-anchored decision rule using subject-cluster bootstrap intervals with positive and negative controls. It was validated on synthetic cohorts of known ground truth and applied to PPG→glucose (three public cohorts, 23+16+24 subjects) and PPG→blood pressure (219 subjects).** ***Results:*** **The block-scan detector found every corrupted subject for cycle lengths $k \geq 3$ with no false alarm on 63 real subjects, and flagged 21/21 patients of a public glucose derivative. Under honest splits the decision rule's false-claim rate stayed at or below the nominal 2.5%; under leaky splits it reached 100%. A 1-D CNN estimated systolic pressure (ΔMAE -2.9$\pm$0.1 mmHg vs. null), yet no model, including frozen PPG foundation models, improved glucose estimation by more than a fraction of the reference's own noise.** ***Conclusion:*** **Leakage-aware, null-anchored evaluation separates genuine physiological signal from artefact.** ***Significance:*** **The audit and its open implementation give wearable-sensing studies a reproducible test for leakage before results are reported.**

***Index Terms*—Photoplethysmography, data leakage, evaluation methodology, blood glucose, blood pressure, cross-validation, wearable sensing, reproducibility.**

I. Introduction

CUFFLESS estimation of blood pressure, glucose and other physiological variables from photoplethysmography (PPG) is an active engineering target, and reported errors often approach those of regulated devices [1]–[3]. Evaluation is the weak link. Record-wise splits of autocorrelated wearable signals inflate accuracy [4]–[6], and recent re-evaluations of PPG-based glucose estimation found that published models do not beat a mean predictor once subjects are separated [7], [8]. Subject-wise splitting is now the recommended remedy [9], [10], but it is not sufficient: leaks introduced while a dataset is *constructed* survive any split, and the magnitude of a leak depends on the model as well as on the protocol.

This paper treats evaluation as an engineering problem with a method that can itself be validated. Its contributions are:

1) **An audit method** (Sec. II) with four components: a detector for label-construction leaks with a block-scan variant for partial corruption, split protocols whose no-leak guarantee is asserted in code before training, a leakage ladder that measures inflation mechanism by mechanism, and a null-anchored decision rule with controls.
2) **Validation on known ground truth** (Sec. III): detector sensitivity and specificity under injected corruption, and the false-claim rate and power of the decision rule under honest and leaky protocols.
3) **Two applications** (Sec. IV). For PPG→glucose the audit uncovers a previously unreported label-replication artefact in a public derivative and bounds what three cohorts support; for PPG→blood pressure it recovers a genuine signal with the same model that fails on glucose, showing the negative glucose result is a property of the target, not of the audit.

TABLE I
Leak levels, the protocol that removes each, and the audit check.

Leak level	Mechanism	Protocol	Audit check
Window	overlapping windows of one record on both sides	record-wise or stricter	set intersection of windows
Record	records of one subject on both sides	subject-wise	set intersection of records
Subject	identity learnt, label = subject mean	subject-wise	set intersection of subjects
Temporal	neighbouring readings across the cut	chronological + guard gap g	min. time distance $> g$
Label construction	labels cycled over signal rows (cross-join)	none (pre-split)	cycle detector, global and block scan

II. Audit Method

A. Setting and leak taxonomy

A dataset is a set of windows w, each with a signal segment, a reference label y, a *record* (one independent reference measurement), a *subject* and, for longitudinal data, a time t. A split protocol produces folds $(\mathcal{T}_f, \mathcal{E}_f)$. We distinguish five leak levels (Table I): window, record, subject and temporal leaks arise from the split; label-construction leaks arise before any split, when a table pairs signal rows with labels so that a label no longer belongs to the signal it accompanies. Kapoor and Narayanan's taxonomy [11] covers the first four; the fifth is closest to their duplicates type but acts on the label column.

B. Label-construction detector

A cross-join of a subject's k reference values with its signal rows produces a label column with $y_i = y_{i-k}$ for almost all

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i, whereas a genuine table is record-structured (runs of one label per record). For a label column $y_{1:n}$ we compute the lag-ℓ match fraction $\phi(\ell) = \frac{1}{n-\ell}\sum_{i>\ell} \mathbb{1}[y_i = y_{i-\ell}]$ for $\ell = 1, \ldots, \ell_{\max}$ and report the smallest ℓ with $\phi(\ell) > \tau$ ($\tau = 0.98$). $\ell = 1$ indicates runs (genuine); $\ell \geq 2$ indicates cycling. Because a corrupted stretch covering a fraction p of a subject's rows is diluted to $\phi(k) \approx p + (1-p)(1-k/m)$ for run length m, the whole-column test misses partial corruption; the *block scan* applies the same test to contiguous blocks of b rows (stride $b/2$). On a record-structured column any block is a single run or crosses a run boundary, so $\phi(1) \geq 1 - 1/b > \tau$ whenever $b > 1/(1-\tau)$ and $b > \ell_{\max}$: the block scan cannot raise a false alarm on genuine record-structured labels.

C. Split protocols with asserted guarantees

Each protocol declares the leak level it removes (Table I). Before any model is trained, the folds are checked by exact set arithmetic on window, record and subject identifiers and, for the temporal level, by the minimum distance between a test reading and any training reading of the same subject. A violation aborts the run. Folds are frozen to files with a fingerprint so a third-party model is scored on identical folds.

D. Leakage ladder

The same model is evaluated under a sequence of protocols that leak progressively less (random windows → random records → subjects; or random readings → chronological → chronological with guard gap). The difference in a metric between adjacent rungs measures the inflation due to one mechanism, separately from the others.

E. Null-anchored decision rule and controls

Window predictions are averaged per record. Every model is compared with the null appropriate to the use case: the training mean for cross-subject estimation; each subject's training mean, a time-of-day model and persistence for within-subject tracking. The paired MAE difference Δ is bootstrapped by resampling *subjects* with replacement (5 000 draws), and a model is declared better than the null only if the 95% interval of Δ lies entirely below zero. One comparison per cohort is pre-specified as primary; the others are exploratory. Positive controls (quantities that must be recoverable, e.g. heart rate) and negative controls (permuted labels, noise input) are run through the identical pipeline, and the reference device's own error (ISO 15197 meter, CGM MARD) is reported as a noise floor.

III. Validation on Known Ground Truth

Detector. We generated label columns for 20 subjects, corrupted a fraction (10–100%) of them with cross-joins of cycle length $k \in \{2, \ldots, 10\}$ over 25, 50 or 100% of their rows, and left the rest record-structured. Fig. 1a shows subject-level sensitivity. Both detectors had 100% specificity in every condition. The whole-column test detected fully corrupted subjects (sensitivity ≥97%) but missed partial corruption for long cycles (minimum 0%); the block scan detected every corrupted subject for $k \geq 3$ (100%) and at least 97% overall, the residual misses at $k = 2$ being draws in which the two reference values coincide and the cycle is invisible. On the label columns of 63 real subjects from three clean cohorts it raised 0 alarms; on the public glucose derivative of Sec. IV-B it flagged 21 of the 21 patients with more than one reference value.

Decision rule. We generated 200 synthetic cohorts per condition (16 subjects, 12 records of three 5-s windows each). Each label is a subject offset (SD 20 mg/dL, unlearnable from the signal) plus g times the heart-rate deviation encoded in the waveform plus reference noise. Each subject also has a pulse-shape fingerprint unrelated to the label, so a flexible model can learn identity, and hence the offset, whenever a protocol lets a subject's windows into training. Fig. 1b shows how often the rule declared a ridge or a gradient-boosting (GBM) model better than the train-mean null. At $g = 0$ under the honest subject-wise protocol the false-claim rate was 1% (ridge) and 1% (GBM), at or below the nominal 2.5%. Under the leaky random-window protocol it rose to 19% and 100%, the GBM *appearing* to improve on the null by -9.3 mg/dL with no signal present. This is why the rule is only valid together with an asserted protocol. With signal present the honest rule's power rose with g, to 96% (ridge) and 93% (GBM) at $g = 4$.

IV. Applications

A. Data and models

Glucose. Kermani–Esmaeili finger PPG [12] (67 10-s recordings, 23 subjects, fingerstick reference); BIG IDEAs wrist BVP with Dexcom G6 CGM [13], [14] (16 subjects, 28867 readings, HbA1c 5.3–6.4%); and STUBA ePPG [15] (125 records, 24 subjects, 3 with type-1 diabetes). **Blood pressure.** PPG-BP [16]: 219 subjects, one cuff reading and three 2.1-s finger-PPG segments each. Signals are band-passed 0.5–8 Hz, resampled to 64 Hz and z-scored per window (5 s for glucose). Models: nulls; ridge and gradient boosting on nine hand-crafted PPG features, with and without demographics; a 1-D CNN, 2-D wavelet-scalogram CNN, Takens-embedding PointNet and their cross-attention fusion [17]–[19]; re-implemented ResNet-1D and CNN–LSTM [2], [20]; and ridge probes on frozen embeddings of three PPG foundation models (PaPaGei-S [21], Pulse-PPG [22], AnyPPG [23]). Networks use one shared configuration (AdamW, fixed budget, no early stopping on held-out data) and three seeds. Protocols: leave-one-subject-out (fingerstick cohorts), chronological 70/30 with a 1-h guard gap (BIG IDEAs; plus a personalised arm trained on the residual to each subject's training mean), and 5-fold subject-wise (PPG-BP). Full tables are in the supplement.

B. PPG → glucose

Label-construction leak. A widely shared Kaggle derivative of the Kermani–Esmaeili recordings [24] cycles each patient's reference values row by row (Fig. 2), which the detector flags for every patient with more than one value. Every window of that file inherits the patient mean, so a per-patient oracle scores MAE 0.03 mg/dL with $r = 1.00$ under

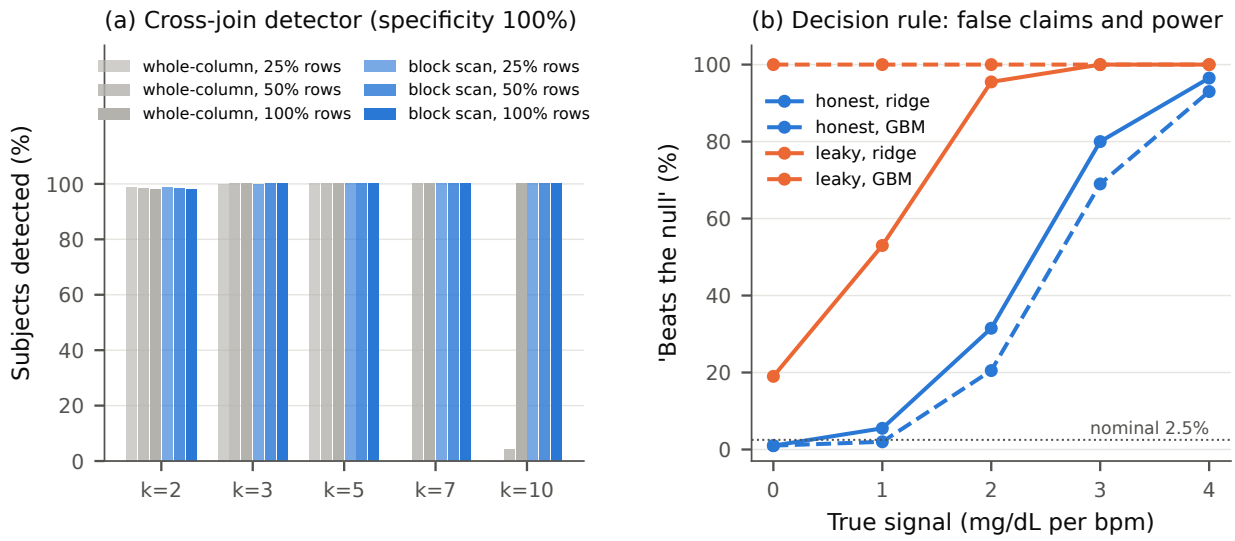


Fig. 1. Validation on known ground truth. (a) Subject-level sensitivity of the cross-join detector, whole-column versus block scan, by cycle length k and fraction of corrupted rows; specificity was 100% throughout. (b) Rate at which the decision rule declares a model better than the null, versus the true learnable signal g (mg/dL per bpm), under the honest subject-wise and the leaky random-window protocol; dotted line: nominal one-sided 2.5%.

a random split, and a correct subject-wise split cannot reveal the problem. An earlier, unpublished draft of this work trained on the file and reported MAE 6.89 mg/dL; those numbers are not carried forward.

Ladder. On the raw recordings the 1-D CNN scores 10.4±0.2 mg/dL ($r = 0.66 \pm 0.05$) under random windows, but 13.5±0.5 and 13.4±0.5 once recordings or subjects are separated, against a null of 14.7 (Fig. 3a): on this cohort window-neighbour leakage, not identity leakage, produces the inflation, because most subjects contribute one or two recordings. On BIG IDEAs the ladder is nearly flat (16.1, 16.9, 16.9 mg/dL) and every rung stays above the subject-mean null.

Bounds. Table II summarises the honest protocols. On the fingerstick cohort the pre-specified primary comparison (1-D CNN) gives Δ -1.4 mg/dL (95% CI [-3.0, -0.03]), nominally significant for 1 of 3 seeds and shrinking to 14.4±0.2 vs. 14.6 under 5-fold splitting. On BIG IDEAs every population model is worse than each subject's own mean; with the personalised arm the best model, a Pulse-PPG probe, reaches Δ -0.3 mg/dL ([-0.5, -0.02]), one of 2 of 41 intervals below zero and an order of magnitude below the CGM noise floor (10.5 mg/dL); median within-subject correlation stays below 0.2 for every model. On STUBA no model beats the null. No model detects any of the 68+34 hypoglycaemic references. Positive controls behave as required: heart rate is recovered from the same windows ($r = 0.74 \pm 0.01$), while permuted labels and noise inputs return to the null.

TABLE II
PPG→GLUCOSE UNDER THE HONEST PROTOCOLS (SELECTED ROWS; ALL 41 COMPARISONS IN THE SUPPLEMENT). MAE IN MG/DL; Δ: PAIRED MAE DIFFERENCE VS. THE NULL WITH 95% SUBJECT-CLUSTER CI. ⋆: PRE-SPECIFIED PRIMARY COMPARISON.

Cohort	Model	MAE	r	Δ [95% CI]
K–E (LOSO)	Train mean (null)	14.7	–	–
	Ridge, PPG features	14.6	-0.03	-0.1 [-0.6, +0.6]
	1-D CNN ⋆	13.3	0.32±0.02	-1.4 [-3.0, -0.03]
	1D+2D+3D fusion	14.7	0.13±0.05	-0.1 [-1.5, +1.2]
	Pulse-PPG probe	14.2	0.14	-0.5 [-1.7, +0.8]
BIG IDEAs (chrono.)	Subject mean (null)	15.3	–	–
	1-D CNN	16.9	0.21	+1.7 [+0.5, +3.1]
	1D+2D+3D fusion	16.7	0.25	+1.4 [+0.2, +2.9]
	1-D CNN, residual	15.0±0.0	0.44±0.00	-0.2 [-0.7, +0.2]
	Pulse-PPG, residual	15.0	0.44	-0.3 [-0.5, -0.02]
STUBA (LOSO)	Train mean (null)	43.5	–	–
	1-D CNN	49.7±1.3	-0.15±0.01	+6.1 [+2.3, +7.6]
	PointNet (Takens)	43.2	-0.13	-0.3 [-1.3, +1.9]
	AnyPPG probe	43.5	0.04	+0.0 [-0.7, +1.1]

C. PPG → blood pressure

The same audit applied to PPG-BP (Table III, Fig. 4) recovers a genuine signal. Under the honest subject-wise split the 1-D CNN estimates systolic pressure with MAE 13.4±0.1 against 16.2 mmHg for the null (Δ -2.9±0.1, CI [-4.2, -1.4]; $r = 0.54 \pm 0.01$), and ridge on PPG features alone also clears the null (Δ -1.1, [-1.7, -0.6]). The architecture that cannot beat the null for glucose therefore detects blood-pressure information in shorter PPG segments under the same protocol and statistics: the glucose result is a property of the target. Diastolic pressure is weaker: only the 1-D CNN clears the null, and narrowly (Δ -0.8±0.1, [-1.4, -0.05]).

The ladder shows that leakage inflation depends on model capacity. A random split over segments places a subject's other segments, and hence its label, in training. Linear ridge cannot memorise subjects and barely changes (13.6 vs. 13.8 mmHg), whereas gradient boosting does: its systolic MAE falls from 14.6 to 9.4 mmHg and its correlation rises from 0.44 to 0.82 (diastolic: 8.2 to 5.5 mmHg). A leak therefore cannot be characterised by the protocol alone; the audit has to measure it for the model being reported.

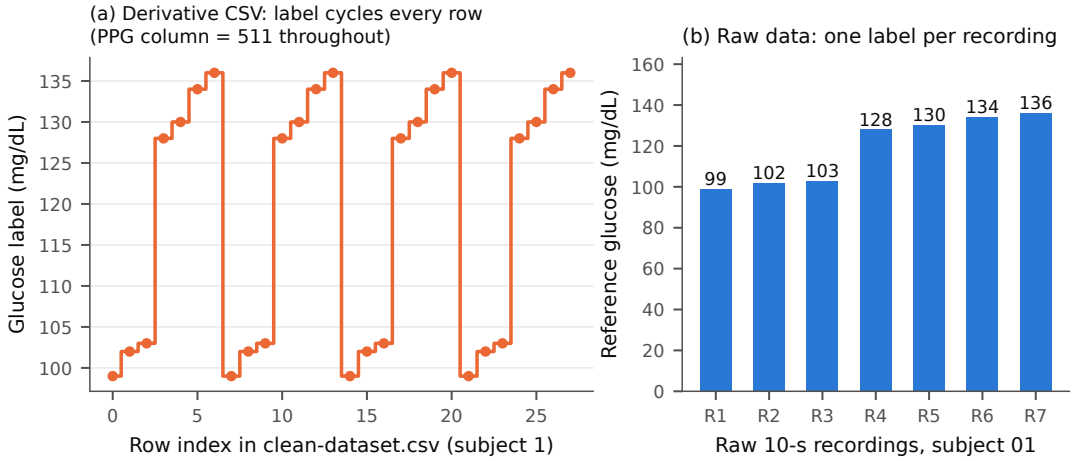


Fig. 2. Label-construction leak in a public glucose derivative. (a) The label of subject 1 cycles through the subject's seven reference values on consecutive rows. (b) The raw recordings of that subject carry one reference each.

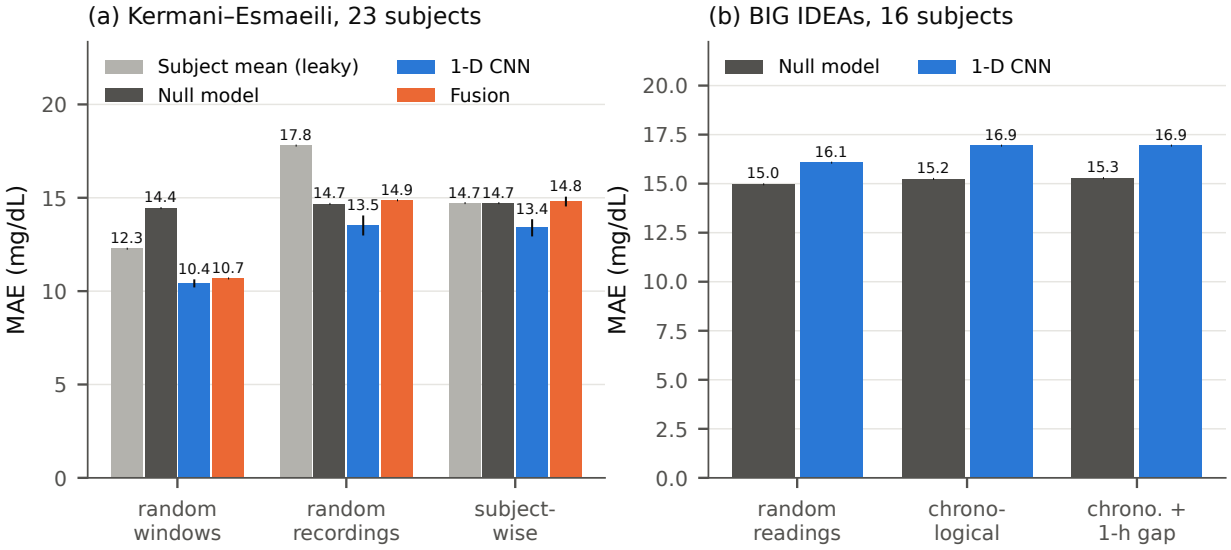


Fig. 3. Leakage ladder for PPG→glucose. (a) Kermani–Esmaeili: random windows, random recordings, subject-wise. (b) BIG IDEAs: random readings, chronological, chronological with 1-h guard gap. Error bars: std over three seeds where available.

V. Discussion

What the audit adds. Subject-wise splitting, the standard remedy, did not detect the largest problem we found: a label-construction leak in a public table, which the detector finds in seconds from the label column alone. The ladder separates mechanisms that are usually conflated: on the glucose recordings window overlap, not identity, inflated the result; on PPG-BP identity leakage mattered only for flexible models. The null-anchored rule held its false-claim rate under honest protocols in simulation and correctly declared the blood-pressure signal while rejecting the glucose claims. Ramesh *et al.* [7] released a re-implementation pipeline with participant-aware splits; the audit differs in asserting each protocol's guarantee in code, detecting construction leaks, measuring inflation per mechanism, and validating its own decision rule.

What the applications show. On public PPG data, a 5–60-s segment carries usable information about systolic blood pressure but, under every architecture tested including frozen foundation models, at most a weak and protocol-dependent cross-subject correlation with glucose and no measurable within-subject tracking; no model detected a hypo- or hyperglycaemic reading. These are statements about the public cohorts, which are small and largely non-diabetic.

Recommendations. Run a construction-leak check on every derived table; name the protocol and the leak it removes, and freeze folds; measure the ladder for the reported model; report a null and the reference noise floor next to every result, with subject-cluster intervals and a pre-specified primary comparison; and include positive controls.

Limitations. The detector targets the cycling form of label construction; other construction errors (e.g. shifted joins) need their own tests. The validation cohorts are synthetic and

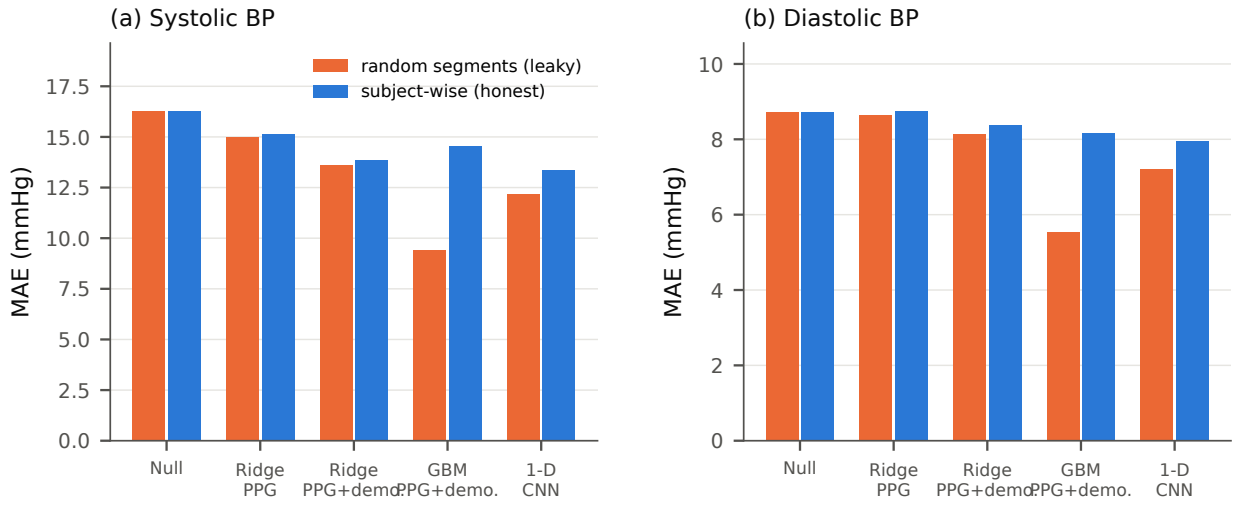


Fig. 4. PPG→blood pressure: MAE of the same models under the leaky random-segment split and the honest subject-wise split. Linear models are barely affected by the leak; flexible models are.

TABLE III
PPG→BLOOD PRESSURE (219 SUBJECTS): MAE (MMHG) UNDER THE LEAKY RANDOM-SEGMENT AND THE HONEST SUBJECT-WISE SPLIT; Δ VS. THE TRAIN-MEAN NULL WITH 95% SUBJECT-CLUSTER CI (HONEST SPLIT; CNN: 3 SEEDS, MOST CONSERVATIVE INTERVAL).

	Model	Leaky	Honest	Δ honest [95% CI]
SBP	Null	16.3	16.2	–
	Ridge, PPG	15.0	15.1	-1.1 [-1.7, -0.6]
	Ridge, PPG+demo.	13.6	13.8	-2.4 [-3.5, -1.3]
	GBM, PPG+demo.	9.4	14.6	-1.7 [-3.1, -0.3]
	1-D CNN	12.2±0.2	13.4±0.1	-2.9±0.1 [-4.2, -1.4]
DBP	Null	8.7	8.7	–
	Ridge, PPG	8.6	8.7	0.0 [-0.2, +0.2]
	Ridge, PPG+demo.	8.1	8.4	-0.4 [-0.8, +0.1]
	GBM, PPG+demo.	5.5	8.2	-0.6 [-1.3, +0.1]
	1-D CNN	7.2±0.1	7.9±0.1	-0.8±0.1 [-1.4, -0.05]

moderate in size. Networks shared one configuration; encoder probes were linear and used one common preprocessing. The glucose cohorts contain few diabetic subjects and few hypoglycaemic references.

VI. Conclusion

We presented and validated an audit for leakage in wearable physiological regression. On known ground truth its detector found label-construction leaks without false alarms and its decision rule held its false-claim rate under honest protocols. Applied to public PPG data it uncovered a label-construction leak in a glucose derivative, recovered genuine blood-pressure information, and bounded the glucose information to a fraction of the reference's own noise. The implementation, frozen folds and all predictions are released so that any new claim can be audited the same way.

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